A Project Report

On

Leveraging Llama 2 For Improved Financial Sentiment Analysis: An Instruction Tuned Approach

Submitted in partial fulfillment of the

requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

GALGOTIAS UNIVERSITY

(Established under Galgotias University Uttar Pradesh Act No. 14 of 2011)

Session 2025-26
in

Computer Science and Engineering

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INDIA

Jan, 2026

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CANDIDATE'S DECLARATION

I/We hereby certify that the work which is being presented in the project, entitled **"Leveraging Llama2 for improved financial sentiment Analyisis -An instruction tuned Approach"** in partial fulfillment of the requirements for the award of the B. Tech. (Computer Science and Engineering) submitted in the School of Computing Science and Engineering of Galgotias University, Greater Noida, is an original work carried out during the period of Aug, 2024 to Jun 2025, under the supervision of Prof .Akhilesh Kumar Singh, Department of Computer Science and Engineering, of School of Computing Science and Engineering , Galgotias University, Greater Noida.

The matter presented in the thesis/project/dissertation has not been submitted by me/us for the award of any other degree of this or any other places.

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CERTIFICATE

This is to certify that Project Report entitled "**Leveraging Llama2 for improved financial sentiment Analyisis -An instruction tuned Approach**" which is submitted by **Disha Singh** and **Ketan Dubey** in partial fulfillment of the requirement for the award of degree B.Tech. in Department of School of Computing Science and Engineering Department of Computer Science and Engineering

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Place: Greater Noida

ACKNOWLEDGEMENT

It gives us a great sense of pleasure to present the report of the B. Tech Project undertaken during M. Tech. Final Year. We owe special debt of gratitude to Professor Akhilesh Kumar Singh, Department of Computer Science & Engineering, Galgotias University, Greater Noida, India *for his constant support and guidance throughout the course of our work. His/Her sincerity, thoroughness and perseverance have been a constant source of inspiration for us. It is only his cognizant efforts that our endeavors have seen light of the day.*

We also take the opportunity to acknowledge the contribution of Professor ***Dr. Harish Kumar G R*** *Dean, Department of Computer Science & Engineering,* Galgotias University, Greater Noida, India *for his full support and assistance during the development of the project.*

We also do not like to miss the opportunity to acknowledge the contribution of all faculty members of the department for their kind assistance and cooperation during the development of our project. Last but not the least, we acknowledge our friends for their contribution in the completion of the project.

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ABSTRACT

The financial sentiment analysis plays a significant role in the formulation of an investment decision, risk management as well as the forecasting of trends in the financial market. It is, however, not an easy mission since it involves the use of jargon, language nuances and succinctness of financial literature. Though the transformer-based FinBERT, ALBERT, and RoBERTa models are satisfactorily performing, they are generally limited by their formulation and performance following the setting-specific statements of finance. Although LLCs, including LLaMA-2, show good overall language comprehension, their characteristics do not seem to be in line with the task of sentiment classification. We offer the instruction-tuned LLaMA-2-7BChat financial sentiment analysis model in this paper. We restructure the Financial PhraseBank dataset of around 2,417 financial statements that have been sentiments of either positive, negative, or neutral into the pairs of instructionresponse in order to more suitably correlate with the task through supervised instructiontuning The proposed model is assessed through typical metrics of classification. It was simply compared to a variety of baseline systems including FinBERT, ALBERT, RoBERTa, and DistilGPT. The LLaMA-2 instruction-tuned, according to the results of the experiments, is at 83.67 percent accuracy and 83.12 percent of F1-score. As a result, this model achieves higher results than any other baselines on the same dataset. These findings depict that instruction tuning generates better understanding of financial sentiment of large language models. It demonstrates how effective the LLaMA-2 can be as a substitute solution to the conventional transformer-based financial sentiment analysis systems.

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CHAPTER1
INTRODUCTION

1.1 Background

In recent years, the rapid growth of digital financial platforms has resulted in an enormous increase in the volume of financial text data. This data includes earnings reports, corporate disclosures, analyst opinions, press releases, and financial news articles. Extracting meaningful insights from such unstructured textual data has become a crucial task in modern financial analytics. Financial sentiment analysis plays an important role in this process by identifying the underlying sentiment expressed in financial texts and categorizing it into predefined classes such as positive, negative, or neutral.Financial sentiment analysis supports a wide range of applications, including investment decision-making, risk management, algorithmic trading, and market trend forecasting. Accurate sentiment interpretation enables stakeholders to respond proactively to market changes and reduce uncertainty in financial planning. However, financial text differs significantly from general-domain text due to the use of specialized terminology, implicit sentiment expressions, and carefully worded neutral statements. These characteristics make financial sentiment analysis more complex and challenging than general sentiment classification.

1.2 Problem Statement

Despite the importance of financial sentiment analysis, accurately identifying sentiment in financial texts remains a difficult problem. Traditional sentiment analysis approaches rely on rule-based systems and classical machine learning models that require extensive feature engineering. These approaches often fail to capture contextual meaning and struggle to adapt to different financial domains.Recent advances in deep learning, particularly transformer-based architectures such as BERT and its variants, have improved sentiment classification performance by modeling contextual word representations. Domain-adapted models such as FinBERT have further enhanced performance by incorporating financial text during pretraining. However, these encoder-based transformer models are primarily designed for discriminative classification tasks and depend on fixed output heads. As a result, they often lack flexibility when handling short, ambiguous, or context-dependent financial statements that require implicit reasoning.Large language models (LLMs), such as LLaMA-2, demonstrate strong general language understanding and reasoning abilities. However, their pretraining objectives are focused on next-token prediction rather than direct task alignment. This mismatch can lead to inconsistent performance when LLMs are directly applied to sentiment classification tasks without additional alignment strategies.

1.3 Motivation

The motivation behind this project is to bridge the gap between the powerful reasoning capabilities of large language models and the specific requirements of financial sentiment analysis. Instead of treating sentiment analysis as a purely discriminative classification task, this work reformulates it as an instruction-following generation problem. By explicitly providing task instructions to the model, it becomes possible to align the model's outputs with the desired sentiment labels more effectively.Instruction tuning has emerged as a promising technique for improving task adherence and output consistency in large language models. Additionally, parameter-efficient fine-tuning techniques such as Low-Rank Adaptation (LoRA) make it feasible to fine-tune large models using limited computational resources. These techniques collectively motivate the development of an instruction-tuned LLaMA-2 framework for financial sentiment analysis.

1.4 Objectives of the Project

The main objectives of this project are as follows:

- To study existing financial sentiment analysis techniques and their limitations
- To reformulate financial sentiment analysis as an instruction-following task
- To design an instruction-tuned framework based on LLaMA-2-7B-Chat
- To apply parameter-efficient fine-tuning using Low-Rank Adaptation
- To evaluate the proposed approach against established baseline models using standard metrics

1.5 Scope of the Project

This project focuses on sentence-level financial sentiment analysis using the Financial PhraseBank dataset. The scope is limited to three sentiment categories: positive, negative, and neutral. The study evaluates the effectiveness of instruction tuning and LoRA-based fine-tuning within a controlled experimental setup. While the proposed approach shows strong performance, the project does not address real-time financial data processing or large-scale industrial deployment.

1.6 Organization of the Report

The remainder of this report is organized as follows:

- **Chapter 2** presents a detailed review of related work and background concepts
- **Chapter 3** describes the proposed methodology and system architecture
- **Chapter 4** explains the experimental setup and performance evaluation
- **Chapter 5** discusses the results and finding

- **Chapter 6** concludes the report and outlines future research directions

CHAPTER 2

LITERATURE REVIEW

2.1 Financial Sentiment Analysis

Financial sentiment analysis focuses on identifying the underlying sentiment expressed in financial text data such as earnings reports, corporate announcements, analyst recommendations, press releases, and financial news articles. Unlike general sentiment analysis, financial sentiment analysis must deal with domain-specific language, carefully phrased statements, and implicit expressions of sentiment. Financial texts often avoid emotional language and instead convey sentiment through indicators such as revenue changes, forecasts, comparative performance, or risk disclosures. As a result, sentiment is frequently subtle and context-dependent.

Accurate financial sentiment classification is critical for various financial applications. Investors rely on sentiment analysis to support investment decisions and portfolio management. Financial institutions use sentiment signals for risk assessment and market trend analysis, while algorithmic trading systems incorporate sentiment features to predict short-term price movements. Incorrect sentiment interpretation can lead to flawed financial judgments and significant economic losses. Consequently, improving the accuracy and reliability of financial sentiment analysis has remained an active research area.Early research in financial sentiment analysis demonstrated that general-purpose sentiment models often perform poorly when applied to financial texts. This limitation arises due to differences in vocabulary, writing style, and semantic structure between financial and general-domain corpora. These challenges motivated the development of domain-aware approaches specifically designed for financial language understanding.

2.2 Evolution of Sentiment Analysis Models

The development of sentiment analysis techniques has progressed through several distinct phases. Early sentiment analysis approaches relied on lexicon-based methods and traditional machine learning algorithms. Lexicon-based techniques used predefined sentiment dictionaries to assign polarity scores to words or phrases. Machine learning approaches employed handcrafted features such as term frequency, n-grams, or syntactic patterns, combined with classifiers like naïve Bayes and support vector machines. While these methods were effective in controlled environments, they required extensive feature engineering and struggled with domain adaptation.

The introduction of deep learning models marked a major advancement in sentiment analysis. Neural architectures such as convolutional neural networks and recurrent neural networks enabled automatic feature extraction from raw text. These models improved sentiment classification by capturing local and sequential patterns in text. However, they faced difficulties in modeling long-range dependencies and often suffered from performance degradation when processing longer or more complex sequences.Transformer-based architectures addressed these limitations by introducing self-attention mechanisms capable of modeling global context. Models such as BERT demonstrated strong performance across a wide range of natural language processing tasks, including sentiment analysis. The ability of transformers to capture bidirectional contextual information significantly improved sentiment understanding and robustness. These advancements laid the foundation for the development of specialized transformer models tailored to specific domains such as finance.

2.3 Transformer-Based Models for Financial Sentiment Analysis

Transformer-based models have become the dominant approach for financial sentiment analysis due to their ability to learn contextual representations of financial language. Encoder-based transformers such as BERT provide deep bidirectional representations that capture subtle semantic relationships. Domain-adapted variants, including FinBERT, further improve performance by pretraining on large-scale financial corpora. This domain-specific pretraining enhances sensitivity to financial terminology, reporting styles, and contextual sentiment cues.Other transformer models such as ALBERT and RoBERTa have also been applied to financial sentiment classification. ALBERT introduces parameter-sharing mechanisms to reduce model size while maintaining performance, making it more computationally efficient. RoBERTa improves training strategies and optimization techniques, resulting in robust representations across various text classification tasks. These models demonstrate strong performance when fine-tuned for financial sentiment classification.Despite their effectiveness, encoder-based transformer models are typically designed as discriminative classifiers with fixed output heads. This design limits flexibility when handling short or ambiguous financial statements that require implicit reasoning or broader contextual understanding. Additionally, adapting these models to new task formulations often requires architectural modifications or extensive retraining.

2.4 Large Language Models and Task Alignment

Large language models represent a shift toward generative, decoder-based architectures trained on massive and diverse text datasets. Models such as GPT-style architectures and LLaMA-2 demonstrate strong general-purpose language understanding, reasoning, and text generation capabilities. Due to their scale and diversity of training data, these models can capture complex semantic relationships that smaller, task-specific models may fail to learn.

However, large language models are primarily trained using next-token prediction objectives. While this training paradigm enables strong generative performance, it does not inherently ensure alignment with downstream classification tasks such as sentiment analysis. When directly applied to classification problems, LLMs may generate verbose, inconsistent, or poorly structured outputs. This issue becomes more pronounced in specialized domains like finance, where precise and reliable predictions are essential.These limitations highlight the importance of task alignment strategies that can guide large language models to produce structured and task-consistent outputs. Without proper alignment, the potential advantages of LLMs may not fully translate to domain-specific sentiment analysis tasks.

2.5 Instruction Tuning in Natural Language Processing

Instruction tuning has emerged as an effective approach for improving task alignment in large language models. In instruction tuning, datasets are reformulated into instruction–response pairs, where each input explicitly describes the task and the expected output format. This training paradigm encourages models to follow instructions accurately rather than relying on implicit associations between inputs and labels.

Research has shown that instruction-tuned models exhibit improved task adherence, better generalization, and increased output consistency across a wide range of natural language processing tasks. Instruction tuning allows large language models to be adapted to new tasks without retraining from scratch, making it computationally efficient. It can be combined with supervised fine-tuning and parameter-efficient methods, further reducing resource requirements.Although instruction tuning has been extensively studied in general NLP settings, its application to financial sentiment analysis remains limited. Few studies have systematically examined how instruction-tuned large language models compare with established financial-domain transformer models. This gap in the literature motivates the present work.

2.6 Research Gap and Motivation for the Proposed Work

Existing financial sentiment analysis models primarily rely on encoder-based transformer architectures optimized for discriminative classification. While these models perform well, they lack the flexibility and reasoning capabilities of large language models. At the same time, large language models have not been sufficiently explored for financial sentiment analysis due to task alignment challenges and computational constraints.This study addresses these limitations by proposing an instruction-tuned framework based on LLaMA-2-7B-Chat. By reformulating financial sentiment analysis as an instruction-following generation task and applying parameter-efficient fine-tuning with LoRA, the proposed approach aims to combine the strengths of large language models with the structured requirements of financial sentiment classification.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 Overview of the Proposed System

This chapter presents the methodology adopted to develop an instruction-tuned large language model for financial sentiment analysis. The proposed system is designed to overcome the limitations of traditional sentiment classification approaches by reframing the task as an instruction-following generation problem. Instead of relying on a fixed classification head, the model is guided through natural language instructions to produce sentiment labels directly.

The methodology integrates three key components: a large pre-trained language model, an instruction-tuning strategy, and a parameter-efficient fine-tuning technique. The base architecture used is LLaMA-2-7B-Chat, a decoder-only transformer model optimized for conversational and instruction-following tasks. To make fine-tuning feasible on limited hardware resources, Low-Rank Adaptation (LoRA) is employed within a Parameter-Efficient Fine-Tuning (PEFT) framework. The Financial PhraseBank dataset is used as the primary data source, which is transformed into an instruction–response format to enable instruction tuning.The overall workflow of the proposed system includes dataset preparation, instruction formulation, model fine-tuning using LoRA, and performance evaluation against established baseline models.

3.2 Dataset Description and Preprocessing

The Financial PhraseBank dataset is used in this study as it is a well-established benchmark for financial sentiment analysis. The dataset consists of approximately 2,417 short financial text passages extracted from news articles, business reports, and corporate financial statements. Each sentence is manually annotated by domain experts and categorized into one of three sentiment classes: positive, negative, or neutral.Due to the importance of preserving financial semantics, preprocessing is deliberately kept minimal. Financial texts often rely on numerical indicators, punctuation, and domain-specific terminology to convey sentiment. Therefore, punctuation marks, numerical values, and financial terms are retained. Text normalization is limited to basic cleaning operations such as whitespace correction and label standardization.To ensure label consistency and data quality, samples with ambiguous or missing annotations are removed. The dataset is then split into training, validation, and test subsets using a fixed split. The training set contains 1,449 samples, while both the validation and test sets contain 484 samples each. This split is applied consistently across all models evaluated in this study to ensure fair comparison and reproducibility.

3.3 Instruction Dataset Construction

A key aspect of the proposed methodology is transforming the standard sentiment classification dataset into an instruction-tuned format. Instead of providing only input text and sentiment labels, each sample is reformulated as an instruction–response pair. In this format, the input consists of a clear natural language instruction followed by the financial sentence, and the output consists of a single sentiment label.

For example, each instruction explicitly asks the model to identify the sentiment of a given financial statement and specifies the expected output format. The response is restricted to one of the predefined sentiment labels: positive, negative, or neutral. This transformation enables the model to learn the task objective explicitly rather than inferring it implicitly from labeled examples.By incorporating task instructions into the input sequence, the model learns to jointly interpret the task definition and the content of the financial text. This approach improves output consistency, reduces ambiguity during inference, and aligns the training objective with the generative nature of large language models.

3.4 Model Architecture

The primary architecture employed in this study is LLaMA-2-7B-Chat-HF. This model is a decoder-only transformer architecture designed specifically for conversational and instruction-following tasks. It consists of multiple stacked self-attention layers and feed-forward neural networks and is trained using a causal language modeling objective.The chat-optimized variant of LLaMA-2 is selected because it demonstrates stronger instruction-following behavior and produces more structured outputs compared to the base model. This makes it particularly suitable for instruction-tuned sentiment classification, where concise and well-formatted responses are required.Input sequences are tokenized using the LLaMA-2 tokenizer. A maximum sequence length of 256 tokens is used. This length is sufficient to capture the full context of financial statements while avoiding excessive padding and unnecessary computational overhead.

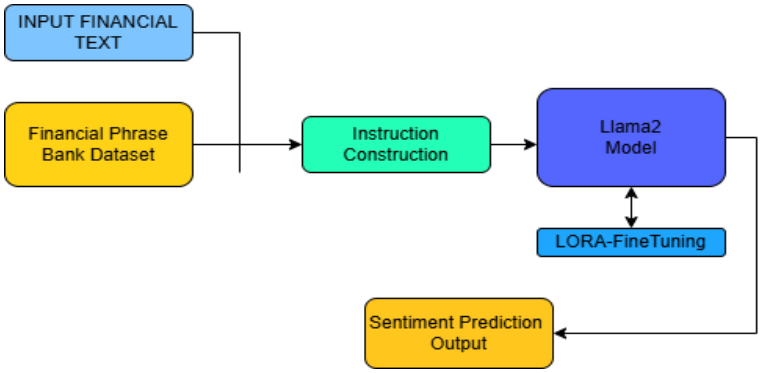


Fig 3.4 workflow of the proposed instruction-tuned LLaMA-2 framework for financial sentiment analysis

3.5 Parameter-Efficient Fine-Tuning with LoRA

Fine-tuning a large language model with billions of parameters is computationally expensive and impractical on limited hardware. To address this challenge, this study employs Low-Rank Adaptation (LoRA) within a Parameter-Efficient Fine-Tuning framework.LoRA introduces additional trainable low-rank matrices into the attention projection layers of the transformer. Instead of updating all model parameters, only these low-rank matrices are trained while the original pretrained weights remain frozen. This significantly reduces memory usage and training time while maintaining strong task performance.In this study, the rank parameter is set to 16, which provides a balance between representational capacity and efficiency. The scaling factor is set to 32 to control the magnitude of the LoRA updates. A dropout rate of 0.05 is applied to reduce the risk of overfitting, especially given the relatively small size of the dataset. This approach enables effective task adaptation without compromising the stability of the pretrained model.

3.6 Training Strategy

Training is performed using a causal language modeling objective. The model is optimized to predict the next token in the instruction–response sequence. During training, the loss is computed using cross-entropy over the target sentiment tokens. This formulation allows the model to generate the correct sentiment label in response to the provided instruction.The AdamW optimizer is used for optimization due to its effectiveness with transformer-based architectures. A learning rate of 1e-4 is selected to ensure stable convergence during fine-tuning while minimizing the risk of catastrophic forgetting of pretrained knowledge. Due to GPU memory limitations, a batch size of one is used. The model is trained for three epochs.All experiments are conducted using the PyTorch deep learning framework in combination with the Hugging Face Transformers and PEFT libraries. Training and evaluation are performed in a Google Colab GPU environment. Model checkpoints are saved at regular intervals, and the best-performing checkpoint is selected based on validation performance.

3.7 Baseline Models

To assess the effectiveness of the proposed instruction-tuned approach, performance is compared against several established baseline models commonly used for financial and general-domain sentiment analysis. These include FinBERT, ALBERT, RoBERTa, and DistilGPT.FinBERT is a domain-specific transformer model pretrained on large-scale financial text. ALBERT and RoBERTa are parameter-efficient and robust transformer architectures that have demonstrated strong performance in sentiment classification tasks. DistilGPT is a lightweight generative language model adapted for classification. All baseline models are fine-tuned using the same dataset splits, preprocessing steps, and evaluation metrics to ensure a fair comparison.

CHAPTER 4

PERFORMANCE EVALUATION AND RESULTS

4.1 Objective of Performance Evaluation

The purpose of this chapter is to present a detailed experimental evaluation of the proposed instruction-tuned LLaMA-2 framework for financial sentiment analysis. The evaluation focuses on measuring how effectively instruction tuning and parameter-efficient fine-tuning improve sentiment classification performance compared to widely used baseline models. This chapter reports quantitative results using standard evaluation metrics and provides a comparative analysis across all models.The experimental evaluation is designed to ensure fairness and reproducibility. All models are trained and evaluated using the same dataset splits, preprocessing steps, and evaluation metrics. This setup guarantees that any observed performance differences are due to model architecture and training strategy rather than experimental bias.

4.2 Dataset Used for Evaluation

All experiments are conducted using the Financial PhraseBank dataset. This dataset consists of 2,417 short financial sentences collected from financial news articles, corporate disclosures, and business reports. Each sentence is manually annotated by financial experts and categorized into one of three sentiment classes: positive, negative, or neutral.

DATASET STATISTICS

Dataset Split	Total Samples	Negative	Neutral	Positive
Training	1449	483	483	483
Validation	484	60	288	136
Test	484	61	287	136
Overall	2417	604	1058	755

Fig 4.2 Dataset Statistics

The dataset is divided into three subsets. The training set contains 1,449 samples and is relatively balanced across sentiment classes. The validation and test sets each contain 484 samples. Unlike the training set, the validation and test sets show a realistic class distribution, with a higher number of neutral samples. This imbalance reflects real-world financial communication and increases the difficulty of sentiment classification.

4.3 Evaluation Metrics

To comprehensively evaluate model performance, four standard classification metrics are used: accuracy, precision, recall, and F1-score.Accuracy measures the proportion of correctly classified samples across all sentiment classes and provides an overall measure of model correctness. Precision measures how many of the predicted sentiment labels are correct, while recall measures how many of the actual sentiment labels are successfully identified by the model. The F1-score represents the harmonic mean of precision and recall, providing a balanced measure that is particularly useful when dealing with class imbalance.For multi-class evaluation, macro-averaged precision, recall, and F1-score are reported. Macro averaging treats all sentiment classes equally, ensuring that positive, negative, and neutral classes contribute equally to the final score.

4.4 Baseline Model Performance

To assess the effectiveness of the proposed model, its performance is compared against four baseline models: FinBERT, ALBERT, RoBERTa, and DistilGPT. Each baseline is fine-tuned using the same dataset splits and evaluated using the same metrics.

4.4.1 FinBERT Performance

FinBERT is a transformer model pretrained on large-scale financial text. On the Financial PhraseBank dataset, FinBERT achieves an accuracy of 0.7872. The model records a precision of 0.7411, a recall of 0.8047, and an F1-score of 0.7650.

These results indicate that FinBERT performs reasonably well in identifying financial sentiment, particularly in recalling sentiment-bearing sentences. However, the lower precision suggests that the model produces a noticeable number of incorrect sentiment predictions, especially in ambiguous cases.

4.4.2 ALBERT Performance

ALBERT is a lightweight transformer model designed to reduce parameter count while maintaining performance. On the evaluation dataset, ALBERT achieves an accuracy of 0.8161. The precision is 0.7760, recall is 0.8399, and the F1-score is 0.7996.Compared to FinBERT, ALBERT shows improved accuracy and F1-score. The higher recall indicates that ALBERT is more effective in identifying sentiment instances, although precision remains moderate, suggesting occasional misclassification.

4.4.3 RoBERTa Performance

RoBERTa is a robust transformer model with optimized training strategies. It achieves an accuracy of 0.8161, which is identical to ALBERT. However, RoBERTa demonstrates better balance between precision and recall. The precision is 0.8398, recall is 0.8161, and the F1-score is 0.8189.Although the accuracy is the same as ALBERT, the higher precision and F1-score indicate that RoBERTa produces more reliable and consistent sentiment

predictions. This highlights the importance of analyzing multiple metrics rather than relying on accuracy alone.

4.4.4 DistilGPT Performance

DistilGPT is a compact generative language model adapted for classification. On the validation set, DistilGPT achieves an overall accuracy of 0.7459. The macro-averaged precision is 0.71, recall is 0.81, and F1-score is 0.73.Class-wise analysis shows that DistilGPT performs well on neutral samples but struggles with precision for negative and positive classes. These results indicate that while generative models can be adapted for sentiment classification, smaller models without task alignment may suffer from inconsistent predictions.

4.5 Proposed Instruction-Tuned LLaMA-2 Performance

The proposed instruction-tuned LLaMA-2-7B model demonstrates the strongest performance across all evaluation metrics. The model achieves an accuracy of 0.8367, which is the highest among all evaluated models. The macro-averaged F1-score is 0.8312, indicating a strong balance between precision and recall across sentiment classes.The improved performance highlights the effectiveness of instruction tuning in aligning the generative capabilities of LLaMA-2 with the financial sentiment classification task. By explicitly framing sentiment analysis as an instruction-following problem, the model produces more consistent and reliable outputs.

MODEL BASED PERFORMANCE ANALYSIS

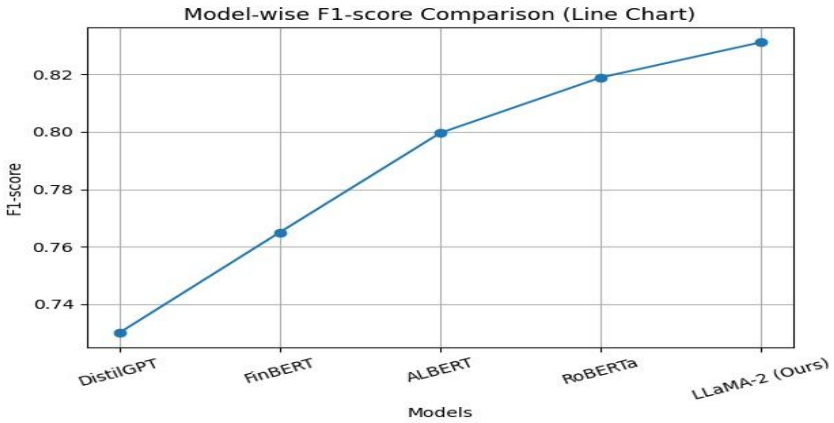
Model	Accuracy	Precision	Recall	F1-score
DistilGPT	0.7459	0.7100	0.8100	0.7300
FinBERT	0.7872	0.7411	0.8047	0.7650
ALBERT	0.8161	0.7760	0.8399	0.7996
RoBERTa	0.8161	0.8398	0.8161	0.8189
LLaMA-2-7B	**0.8367**	**0.8342**	**0.8285**	**0.8312**

Fig 4.5 Model based performance Analysis

4.6 Comparative Performance Analysis

A comparison across all models reveals a clear performance hierarchy. DistilGPT records the lowest accuracy and F1-score, followed by FinBERT. ALBERT and RoBERTa demonstrate competitive performance, with RoBERTa showing better balance between precision and recall. The instruction-tuned LLaMA-2 model outperforms all baseline models in both accuracy and macro-averaged F1-score.Notably, although ALBERT and RoBERTa achieve identical accuracy values, their F1-scores differ. This demonstrates that accuracy alone does not fully capture model performance, especially in imbalanced

datasets. The superior F1-score of LLaMA-2 confirms its robustness across sentiment classes.



4.7 Performance on Neutral and Ambiguous Sentences

Neutral financial statements pose a significant challenge for sentiment analysis due to their implicit and cautious language. The instruction-tuned LLaMA-2 model shows improved handling of neutral samples compared to baseline models. This improvement can be attributed to the model's ability to leverage contextual reasoning and instruction-based task understanding.By treating sentiment classification as a language understanding task rather than a rigid classification problem, the proposed approach captures subtle sentiment cues more effectively.

CHAPTER 5

DISCUSSION

The experimental results demonstrate that instruction tuning significantly enhances the performance of large language models for financial sentiment analysis. The proposed instruction-tuned LLaMA-2-7B model consistently outperforms all baseline models in terms of accuracy and macro-averaged F1-score. This improvement indicates that explicitly aligning the model with task-specific instructions enables more effective interpretation of financial text compared to traditional fine-tuning strategies.A key observation from the comparative analysis is that the instruction-tuned LLaMA-2 model surpasses FinBERT, despite FinBERT being pretrained on large financial corpora. This suggests that task alignment through instruction tuning can be more impactful than domain-specific pretraining alone. By reframing sentiment classification as an instruction-following task, the proposed approach allows the model to leverage its broad pretraining to better capture implicit sentiment cues and contextual nuances present in financial statements.Analysis of precision, recall, and F1-score reveals that while baseline models such as ALBERT and RoBERTa achieve competitive accuracy, they

exhibit trade-offs between precision and recall. In contrast, the instruction-tuned LLaMA-2 model maintains a better balance across these metrics, resulting in the highest macro-averaged F1-score. This balanced performance confirms the robustness of the proposed framework across all sentiment classes.Neutral financial statements remain particularly challenging due to their implicit and cautious language. The proposed model shows improved handling of such cases, which can be attributed to its generative reasoning ability combined with instruction-based task understanding. This enables more accurate interpretation of subtle sentiment indicators that are often missed by discriminative classifiers.The use of Low-Rank Adaptation further contributes to the effectiveness of the proposed approach. Despite limited computational resources and a relatively small dataset, parameter-efficient fine-tuning allows effective model adaptation without overfitting. Overall, the results confirm that instruction tuning combined with parameter-efficient fine-tuning provides a powerful and practical solution for financial sentiment analysis.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project presented an instruction-tuned large language model framework for financial sentiment analysis. The primary objective was to investigate whether instruction tuning can effectively align the reasoning and generative capabilities of large language models with the structured requirements of financial sentiment classification. To achieve this, sentiment analysis was reformulated as an instruction-following generation task using the LLaMA-2-7B-Chat model, combined with parameter-efficient fine-tuning through Low-Rank Adaptation.

The proposed approach was evaluated using the Financial PhraseBank dataset, a widely recognized benchmark for financial sentiment analysis. Experimental results demonstrated that the instruction-tuned LLaMA-2 model outperformed established baseline models, including FinBERT, ALBERT, RoBERTa, and DistilGPT, across standard evaluation metrics such as accuracy and macro-averaged F1-score. These results confirm that explicit task alignment through instruction tuning significantly improves model performance compared to traditional discriminative fine-tuning strategies.Furthermore, the use of parameter-efficient fine-tuning proved to be both effective and practical. By updating only a small subset of trainable parameters, the proposed framework achieved strong performance while maintaining low computational cost. This demonstrates that large language models can be successfully adapted for domain-specific tasks such as financial sentiment analysis even under limited hardware

resources.Overall, this study highlights the potential of instruction-tuned large language models as a powerful alternative to conventional transformer-based sentiment classifiers in financial natural language processing.

6.2 Future Work

While the proposed framework achieved promising results, several directions for future research can further enhance its effectiveness and applicability. One potential extension is the use of larger and more diverse financial datasets, which could improve generalization across different financial domains and writing styles. Exploring multilingual financial sentiment analysis using instruction-tuned models is another valuable direction, particularly for global financial markets.

Future work could also investigate more advanced instruction designs, including multi-step instructions and multi-task instruction tuning, to further improve model reasoning and adaptability. Integrating the proposed approach with real-time financial data streams and market analysis systems would increase its practical relevance for industry applications.Additionally, future research may explore alternative parameter-efficient fine-tuning techniques and optimization strategies to further reduce training cost while maintaining high performance. These extensions would help advance the use of instruction-tuned large language models in real-world financial decision-making systems.

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